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0.6-1 | Eigenvalues and Vectors

Analyze a 3-story shear frame using the flexibility method to determine natural frequencies and mode shapes (after Clough and Penzien [0.6.1]) The model can be used to analyze a two story isolated structure by modeling the first story stiffness and mass as the isolated foundation system. In this example, the model is calibrated to reproduce the Clough example.

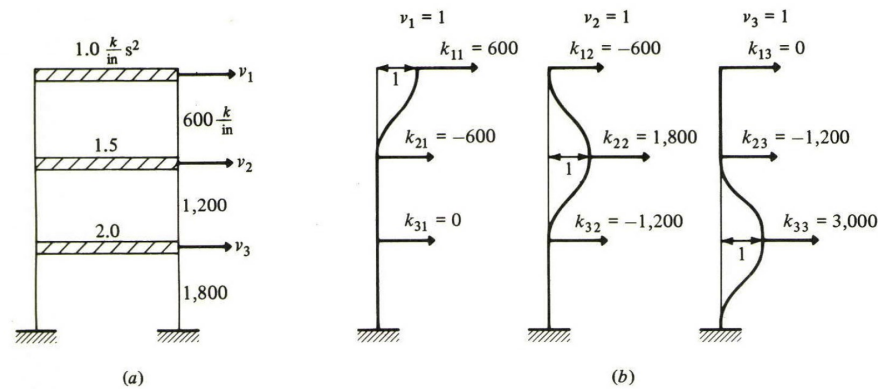


Fig. 1.1 - Structural Model

Normalized mode shapes are calculated and compared to Penzien and Clough. [0.6.2]

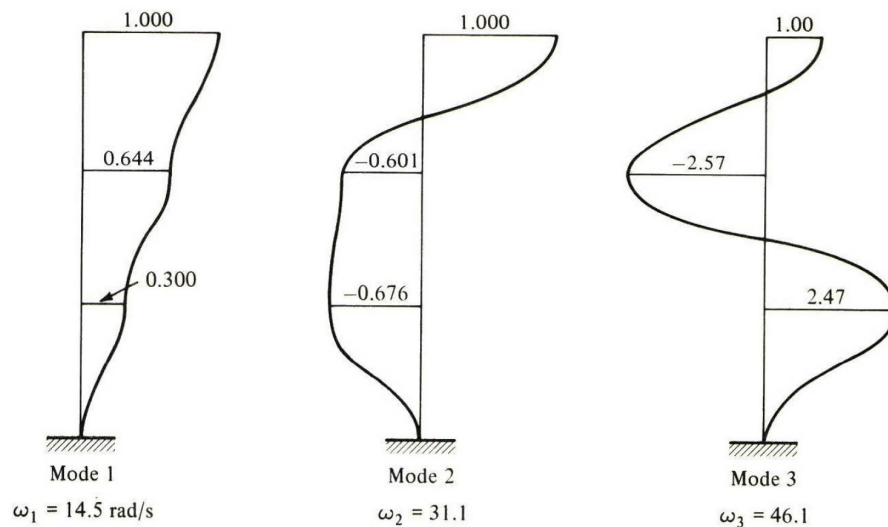


Fig. 1.2 - Mode Shapes from Clough

0.6-2 | Eigen Script

```
import numpy as np
import numpy.linalg as la
import textwrap as tw
# set up mass and stiffness arrays
m = np.array([[1.0,0,0],[0,1.5,0],[0,0,2.0]],float)
k1 = 600*np.array([[1,-1,0.0],[-1,3,-2],[0,-2,5]],float)
# flexibility and dynamic matrix
f = la.inv(k1)
d = np.inner(f,m)
eigen = la.eig(d)
evalus = eigen[0]
#
#                               eigenvalues
print(" "*8,evalus)
      [0.00474206  0.00103739  0.00047055]
#
nat_freq = 1/(np.sqrt(evalus))
#
#                               natural frequencies
print(" "*8,nat_freq)
      [14.52166783  31.04769646  46.09947622]
#
evect = np.array(eigen[1])
#
#                               eigenvectors
print(tw.indent(str(evect), " "*8))
      [[-0.81332769 -0.73942881  0.27304451]
       [-0.52747169  0.44853685 -0.69406171]
       [-0.24550292  0.50205551  0.66612689]]
```



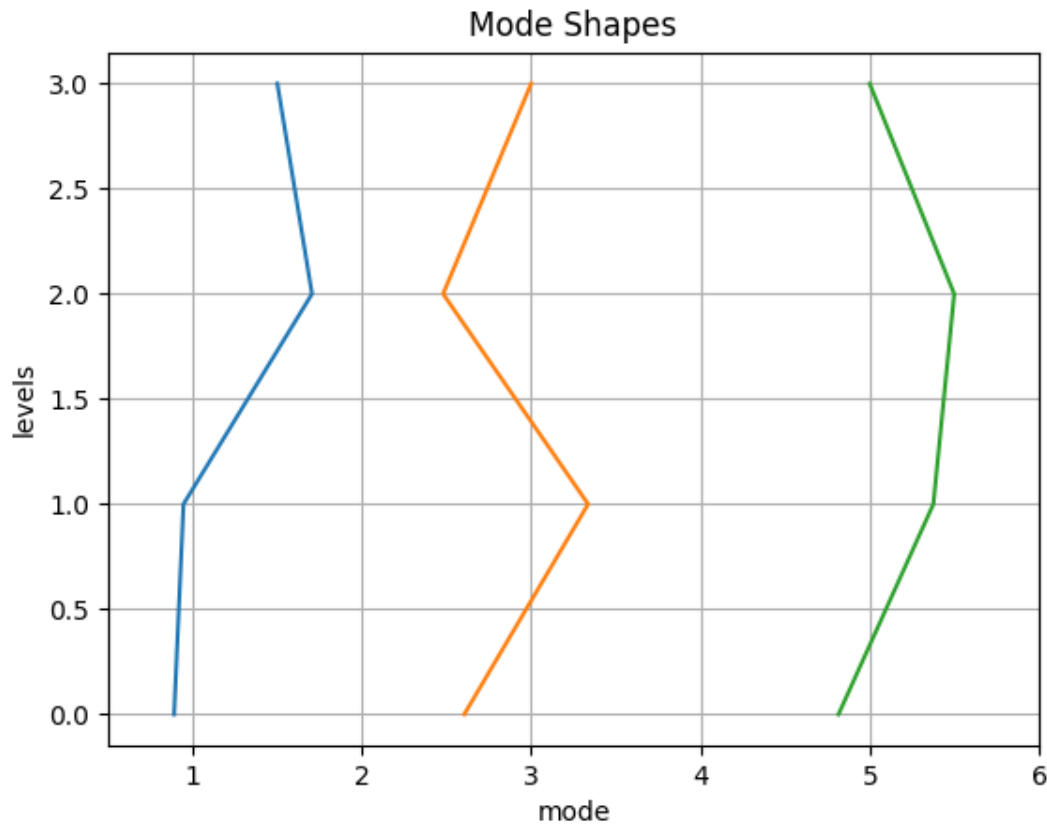
0.6-3 | Plot Script

```

import numpy as np
import matplotlib.pyplot as plt
import os
from tabulate import tabulate as tb
# initialize eigenvector array (need (x,1) shapes for plotting)
ms = np.shape(evect)
zz = np.zeros((ms[0],1))
x1=np.concatenate((evect,zz),1)
# plot mode shapes using matplotlib
y=np.array([0,1,2,3])
m3=x1[2]*.75+5
m2=x1[1]*.75+3
m1=x1[0]*.75+1.5
m=np.concatenate((m1,m2,m3))
plt.clf()
plt.plot(m1,y)
plt.plot(m2,y)
plt.plot(m3,y)
plt.xlim(.5,6.)
plt.xlabel("mode")
plt.ylabel("levels")
plt.title("Mode Shapes")
plt.grid()
curdir=os.getcwd()
imgdir=os.path.join(curdir,"image","mode_shapes.png")
plt.savefig(imgdir)
# table of eigenvalues and normalized eigenvectors
evectt = np.transpose(evect)
for il in range(len(nat_freq)):
    evectt[il] = evectt[il]/evectt[il][0]
v1 = np.concatenate((nat_freq[:, np.newaxis],evect),1)
v2 = np.round(v1, 4)
h1 = ["freq","level 3","level 2","level 1"]
tb1 = np.vstack((h1,v2))
# Table of eigenvalues and normalized eigenvectors
print(tb(tb1, headers="firstrow", tablefmt="rst"))
=====
   freq   level 3   level 2   level 1
=====
14.5217      1      1      1
31.0477    0.6485   -0.6066  -2.5419
46.0995    0.3018   -0.679   2.4396
=====

```



0.6-4 | Plot Mode Shapes**blue:** mode 1**red:** mode 2**green:** mode 3**Fig. 4.1** - Calculated Normalized Modes**[0.6.1]**

R.W. Clough and J. Penzien, Dynamics of Structures. New York, NY, USA:McGraw-Hill, 1975. pg. 178-180

[0.6.2]

ibid. pg. 180-182

